S3 Supplementary Quality Control and Robustness

Export-contract audit and sensitivity checks

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Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

S3.1 Contract and coverage audit

The frozen run manifest and wrapper audit both identified 48 H5 files, 13 unique children, and 11 complete four-session children. All recordings had valid cohort-anchored 3D coordinates and 3D Jacobians; 38 geometry contracts were `ok` and 10 were `adjusted`. The 9D Jacobian was available in 46/48 sessions. `sub-10/ses-4` retained zero 9D Jacobian windows after 140 invalid windows, and `sub-13/ses-2` retained zero after 155 invalid windows; they are excluded only from 9D-Jacobian analyses. This supports computation of the exported session summaries, but does not validate a biological or developmental interpretation of them.

The H5 contract contains subject- and cohort-anchored 3D/9D layers. The common cohort scale uses anchor `ds003710_global_cohort_all_timepoints_iqr_v2_1` (hash `cb876d...d7a6fd7`), subject-balanced over 8,179 epochs. Cohort-anchored coordinate comparisons are therefore valid descriptive comparisons within this export; because the anchor includes all study ages, they are not held-out-age or predictive claims.

S3.2 Session-level aggregation

The wrapper writes exactly one metric row per child-session. Epochs, windows, and local-neighborhood fits affect the within-session median only. The test–retest and longitudinal statistics consequently use 13 and 11 independent child rows, respectively, rather than treating thousands of windows as independent observations.

S3.3 Reachability configuration

The frozen configuration was:

```
coordinate contract: cohort_anchored
space: 3d
k: 12
horizon: 3
estimator: ridge (alpha 0.001)
aggregation: median
```

This fixed configuration guards against choosing a k or horizon after inspecting outcome contrasts. It does not establish that other local-scale settings would have the same behavior.

S3.4 Bootstrap and retest baseline sensitivity

ICC uncertainty used 2,000 bootstrap draws that resampled children as clusters. The primary four-month-mean contrasts are not labelled as sensitivity analyses. The stored retest-mean output applies one pooled 18-test BH family to those same nine-endpoint session 3 and session 4 contrasts and produced no FDR-significant result (minimum adjusted p=0.618). Session-1-only and Friedman outputs are the secondary baseline sensitivities; the nominal Friedman result for `tube_d_eff` was raw p=0.020 and nine-test FDR p=0.177, with no session-1 contrast FDR-supported.

S3.5 Exploratory event analysis

The event wrapper reads the H5 `event_windows` mapping for object, water, speech, and face labels. It computes per-event contrasts against the `pre_near` bin, averages them across available sessions within child, and performs a 2,000-draw sign-flip test. The 12 event-by-axis tests use a separate family. The water-deviation contrast had raw p=0.0055 and BH p=0.066. This output is separated from the primary reliability family; ages are pooled and session counts differ. Three low-label sessions (`sub-08/ses-2`, 0.5%; `sub-13/ses-2`, 0.6%; and `sub-07/ses-2`, 2.5%) contributed no usable event/pre-near pair to the exported tests; the minimum stage-label fraction among contributing event sessions was 15.3%. Any nominal event effect is therefore only a prompt for a confirmatory study.

S3.6 Machine-readable outputs

`01_session_metrics/session_metrics.csv` contains the session endpoints; `session_qc.csv` contains the contract/QC audit; `02_reliability_longitudinal/` contains all reliability and paired-contrast estimates; and `03_event_locked_exploratory/` contains the optional event analysis. The `export_manifest.json` records input path, exact cohort counts, contract, and reachability settings.